

KUNDAN MAHADEV AYARE

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Professional Summary

Engineer with strong project experience in cloud-native DevOps (Kubernetes, Terraform, Argo CD, Docker, GitOps, CI/CD) and applied machine learning (Python, Scikit-learn, clustering, forecasting), backed by professional experience in enterprise software quality, automation, and Agile delivery at EY GDS. Skilled in building self-service developer platforms, GitOps deployment pipelines, observability stacks, and data-driven analysis. Actively growing toward DevOps, Cloud, and ML engineering roles, with hands-on infrastructure automation and applied ML project work.

Technical Skills

DevOps & Cloud: Kubernetes, Docker, Terraform, Argo CD, GitOps, GitHub Actions, CI/CD, Prometheus, Grafana, Loki

Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib, Machine Learning, NetCDF

Languages: Python, Rust, Java, SQL

Frameworks & Tools: Axum, Tokio, REST APIs, Git, Linux, Cosign, Trivy

Testing & QA: Selenium WebDriver, Test Automation, Defect Analysis, System Testing, Agile/Scrum

Education

SRM Institute of Science & Technology, Chennai, India

Sep 2021 – Jun 2025

Bachelor of Technology in Computer Science

Projects

AstraForge – Internal Developer Platform | *Rust, Kubernetes, Argo CD, Terraform, Docker*

- Built a self-service Internal Developer Platform enabling developers to scaffold and deploy microservices via a Rust CLI, cutting service setup from hours to a single command.
- Designed a GitOps-driven deployment pipeline using Argo CD and Kustomize, with GitHub Actions handling lint, test, security scanning (Trivy), image signing (Cosign), and GHCR publishing.
- Provisioned platform infrastructure with Terraform, including a local kind Kubernetes cluster, ingress, cert-manager, Sealed/External Secrets, and full observability via Prometheus, Grafana, and Loki.

Real-Time System Monitor | *Rust, Axum, Tokio, Sysinfo, Docker, Render*

- Developed a real-time system monitoring dashboard in **Rust** using the Axum web framework and Tokio asynchronous runtime.
- Built REST API endpoints to monitor CPU, memory, disk, network, uptime, and system information using the **sysinfo** crate.
- Designed a responsive web dashboard with live system metrics, light/dark theme support, containerized the application using Docker, and deployed it on Render.

Rainfall Clustering Analysis | *Python, NetCDF Data*

- Utilized unsupervised machine learning and clustering techniques to analyze rainfall data, identifying key trends and improving weather forecasting accuracy.
- Collaborated with Dr. Rajib Chattopadhyay (Scientist E) and Senior Research Fellow Ms. Lekshmi to enhance rainfall forecasting algorithms, contributing to a 20% reduction in the misinterpretation of critical atmospheric data.
- Developed data analysis and machine learning workflows using Python, Pandas, NumPy, and Scikit-learn.

Experience

Associate Test Engineer

Mar 2025 – Present

EY GDS

Chennai, India

- Developed automation solutions using Java and Selenium WebDriver, reducing manual testing effort and improving release quality.
- Performed end-to-end validation, defect analysis, and system testing for enterprise applications in Agile development environments.

Research Intern

Jun 2023 – Jul 2023

Indian Meteorological Department

Pune, India

- Collaborated with senior researchers to optimize rainfall forecasting algorithms, reducing misinterpretation of critical atmospheric data by over 20%.
- Performed detailed analyses on vast amounts of atmospheric data highlighted during the internship, providing findings that corrected inaccuracies observed in previous rainfall forecasts shared with senior researchers.

Leadership Experience

Aaruush (National Level Techno-Management Fest)

Committee Head, External Affairs

- Coordinated logistics and cross-functional teams across multiple events for a national-level techno-management fest, ensuring smooth execution.